

Coding & Solution

Date	10 July 2026
Team ID	xxxxxx
Project Name	Raising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[https://github.com/reshmashaik-r11/flood-prediction-app]
Programming Language(s)	Python 3.9+
Framework(s) Used	Flask 3.0, Scikit-Learn 1.3, joblib
Key Features Implemented	Local data logging to CSV pipeline. Real-time pre-trained .pkl model inference for risk classification. Multi-tier color-coded warning dashboard UI.
Pending / Incomplete Features	Automated external meteorological sensor data ingestion interface.
Setup / Run Instructions	Install dependencies from requirements.txt. Run python app.py from the terminal. Access the application dashboard at

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

This local deployment solution prioritizes offline operational readiness for critical early warning systems in disaster management contexts.